Efficient and Lightweight Semantic Segmentation Network for Land Cover Point Cloud With Local–Global Feature Fusion

Hong Hu, Lantian Cai, Ruihong Kang, Yanlan Wu, and Chunlin Wang

***Abstract*— Utilizing deep learning techniques to extract high-precision features from point clouds is essential for accurately capturing land cover information, which is instrumental in urban planning and environmental conservation. Despite delivering high-accuracy outcomes in the semantic segmentation of extensive terrestrial point clouds, prevalent methodologies encounter considerable hurdles, particularly in training and inference duration, as well as the associated hardware expenses. To solve these issues, this article introduces an efficient and lightweight deep learning network called Uniform Voxelization Geometric Enhancement and Local–Global Feature Fusion Network (VEF-Net). VEF-Net is designed to improve the training and inference efficiency of large-scale point cloud semantic segmentation while maintaining accuracy. Uniform voxel downsampling is employed to discretize point clouds, resulting in a substantial enhancement in computational and memory performance. To counteract potential information loss due to voxel downsampling, VEF-Net integrates a mechanism unit to enhance local geometric features, enriching point cloud data. Furthermore, it incorporates a local–global feature fusion module, adeptly capturing the global contextual relationships within the point cloud. Experimental results show that VEF-Net achieved excellent performance on both the proprietary Bengbu dataset and the publicly available Toronto-3D dataset. VEF-Net maintained comparable segmentation accuracy to mainstream models while operating at a lower computational cost, and it demonstrated significant advantages in certain key tasks. On the Toronto-3D dataset, VEF-Net achieved an intersection over union (IoU) of 20.4% in the road marking category. Additionally, VEF-Net demonstrated lower model parameters and computational complexity [floating point operations per seconds (FLOPs)], achieving training speeds 12 times faster and inference speeds 16.8 times faster than point transformer (PT).**

***Index Terms*— Feature fusion, land cover semantic segmentation, point cloud, VEF-Net.**

I. Introduction

LAND cover classification plays a crucial role in various fields, including urban planning, environmental protection, disaster management, and socioeconomic development [1], [2]. Traditionally, land cover analysis has relied on remote sensing imagery sourced from satellite or aerial photography [3], [4]. While these methods have proven effective, they often fall short in capturing vertical structures and fine spatial details [5]. Three-dimensional laser point cloud technology provides high-precision spatial information, overcoming the vertical spatial resolution limitations of images [6]. Unlike remote sensing images, the unstructured and irregular nature of point clouds makes them more challenging to process [7]. Some previous studies have projected point clouds into planar images to extract features using image convolutional neural networks (CNNs) [8], [9]. Concurrently, other investigations have focused on converting point clouds into voxels and applying 3-D convolution for processing [10], [11]. However, these conversion methods may result in the loss of spatial structural information, which can impact the precision of semantic segmentation. Recent advancements, exemplified by PointNet [12] and PointNet++ [13], have introduced direct semantic segmentation of point clouds, circumventing information loss during the conversion process. Nonetheless, these methodologies exhibit notable constraints in capturing intricate details, managing large-scale datasets, and interpreting complex scenes [14], [15].

The advent of the self-attention mechanism has ushered in significant strides in the comprehension of point cloud data [16], [17], [18]. Researchers have developed multiheaded self-attention feature extraction layers, enhancing the granularity of detail captured from point clouds and the depth of understanding in complex scene analysis, which in turn has elevated the accuracy of segmentation tasks [19], [20], [21]. Yet, the self-attention mechanism for point clouds still faces issues of model complexity and memory overhead, making it difficult to apply to large-scale land cover point clouds. To mitigate these challenges, RandLA-Net [22] improves processing efficiency through random sampling and local feature aggregation. Despite its advantages, RandLA-Net's application is often hindered by the need for intricate preprocessing steps. On the other hand, fast point transformers (PTs) [23] efficiently handle large-scale point clouds through local self-attention mechanisms and voxel hashing architecture. However, this method demands the entire point cloud to be loaded into memory, which can be prohibitive in terms of hardware requirements and memory usage. With the proliferation of edge computing and mobile devices, an increasing number of deep learning applications need to operate in environ-

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ments with limited computational resources and storage space, especially in fields such as urban planning, environmental protection, disaster management, and socioeconomic development. In these areas, point cloud data are widely used for modeling and analyzing urban infrastructure, environmental changes, and postdisaster assessments. However, due to the large volume of point cloud data and high-computational demands, traditional methods face significant challenges on resource-constrained devices. Particularly in disaster management, such as after earthquakes, floods, and other calamities, how to quickly assess changes in disaster-stricken areas and provide decision support remains an urgent issue. Therefore, the semantic segmentation of large-scale land cover point clouds through deep learning continues to grapple with the complexities of model design, the demands for advanced computational resources, and the practical considerations of data processing at scale. These challenges underscore the need for innovative solutions that can balance the competing demands of accuracy, efficiency, and scalability in the realm of land cover classification.

In pursuit of augmenting the operational efficiency and practical applicability of large-scale point cloud segmentation models, the Uniform Voxelization Geometric Enhancement and Local–Global Feature Fusion Network (VEF-Net) network is presented in this article. This novel network is meticulously engineered to deliver high-performance segmentation while ensuring streamlined training and inference processes. VEF-Net first uses a uniform grid downsampling module to convert continuous point cloud data into structured grid point clouds, which significantly reduces the computational complexity in subsequent processing stages. Subsequently, 1-D convolution is applied for feature extraction from the point cloud, establishing a robust foundation for expedited model training across the network. To compensate for the loss of point cloud information after uniform voxelization, VEF-Net incorporates a K-nearest neighbors (KNNs) algorithm to identify the K closest neighbors for each point. This strategic approach allows for the computation of localized geometric features, thereby enhancing the richness of the feature set and facilitating a more nuanced extraction of local structural attributes for individual points. Moreover, the feature fusion phase of VEF-Net adeptly amalgamates local and global data insights. It employs a local multilayer perceptron (MLP) feature fusion mechanism alongside a global self-attention mechanism. This dual approach not only fortifies the integration of neighboring point features but also adeptly captures and assimilates global contextual information within the point cloud. By merging local details with a broader global perspective, VEF-Net provides a comprehensive and meticulous feature learning mechanism for the semantic segmentation of point cloud data, significantly amplifying the accuracy of segmentation tasks. VEF-Net has undergone rigorous validation on both the proprietary Bengbu dataset and the publicly accessible Campus3D dataset. The results show that VEF-Net performs well on both datasets, validating its effectiveness and applicability to different types and scales of point cloud data. In summary, our main contributions are as follows.

1) The introduction of VEF-Net, an efficient and lightweight semantic segmentation network for land cover point clouds, which significantly bolsters the efficiency of segmentation tasks while preserving essential accuracy metrics.
2) The development of a local–global feature fusion module, which empowers the network to assimilate local feature nuances and simultaneously consider the overarching global context, thereby enriching the network's feature learning and segmentation capabilities.

II. Related Work

A. Image-Based and Voxel-Based Methods

Image-based and Voxel-based methods facilitate feature extraction by transforming point cloud data into alternative formats. Image-based methods convert 3-D point clouds into multiple views (such as depth maps, red, green and blue (RGB) images, or normal maps), describing the point cloud information from different angles and extracting features from these views. These features capture the geometric structure and appearance of the point cloud, enabling a comprehensive understanding of the point cloud [8], [24], [25]. Multi-view convolutional neural network (MVCNN) [26] projects point clouds from multiple angles into multiple views and uses max pooling to aggregate view features for segmentation, but this may lead to partial information loss. To address this issue, Yu et al. [9] implemented relational networks to extract internal relationships between different views, along with enhancement and integration blocks, to improve the aggregation of view features and thereby increase recognition accuracy.

Voxel-based methods discretize point clouds into voxel grids and use 3-D CNNs to process the voxelized data, extracting spatial structure and geometric features. This approach effectively captures the 3-D information of point clouds, enabling their understanding [10], [27], [28]. Huang and You [10] divided the point cloud into occupied voxels and used 3-D convolution to complete the segmentation, assigning the same semantic label to points within the same voxel. VoxNet [28] maps point cloud data to a regular grid and process it with 3-D CNN for rapid 3-D object recognition. PointGrid [11] combines point cloud and voxel grid representations by sampling a fixed number of points in each grid cell. It uses 3-D convolution to efficiently extract local geometric features, allowing for effective handling of point cloud data.

However, image-based methods may lose structural information during data conversion, and voxel-based methods can increase computational costs and memory usage, especially when dealing with high-resolution data.

B. Point-Based Methods

Point-based methods process raw point cloud data directly, eliminating the need for preliminary data conversion [29], [30]. Qi et al. [12] proposed PointNet, a pioneering approach that ingests raw point cloud data into a neural network. It extracts features using MLPs and employs symmetric functions to achieve point permutation invariance for global information aggregation. The subsequent PointNet++ [13] incorporated hierarchical and local region processing mechanisms to overcome PointNet's limitations in capturing local

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structural details, thereby enhancing the granularity of local feature capture. PointCNN [31] introduced a novel convolutional processing technique for point clouds through X-Conv operations. X-Conv facilitates robust feature extraction from unordered point clouds by learning the arrangement of local point sets, preserving the structural benefits of conventional CNNs. This approach has shown remarkable results in point cloud classification, segmentation, and other tasks, adeptly tackling the challenges associated with processing unordered point cloud data. Deep neural network with spatial pooling (DNNSP) [32] enhances point cloud classification by directly processing point clouds. It generates feature descriptors for each cluster, uses minimum spanning tree-based pooling to capture spatial relationships, and applies max pooling before classification with an MLP. Multi-scale voxel-point adaptive fusion network (MVP-Net) [33] enhanced feature representation across different scales through multiscale voxel fusion modules and geometric self-attention modules, combined with a pyramid decoder.

C. Attention-Based Methods

Attention mechanisms, which have achieved remarkable success in natural language processing, have been increasingly adopted across various domains. Dense connection-based kernel point network (DenseKPNet) [34] expands the receptive field of kernel point convolution for effective semantic and geometric feature extraction. It uses multiscale convolution kernels and dense connections to capture contextual information, while the kernel point convolution attention module (KPCAM) improves feature discriminability. In the realm of point cloud semantic segmentation, AttAN [35] designed an adversarial learning additive attention module that automatically focuses on the crucial parts of multiscale features for segmentation tasks. The PT [19] effectively captures local structures and global contexts of point cloud data through self-attention mechanisms. This capability allows the network to more precisely interpret and manipulate complex geometric relationships within 3-D space. Point cloud transformer (PCT) [20] first reduces the number of point clouds through multiple downsampling steps and then extracts features through stacked multihead attention layers for classification and segmentation tasks. The random sampling point transformer (RSPT) network [21] significantly improves segmentation accuracy by combining a local feature aggregation module with self-attention and residual dense (ResiDense) connections. Fast PT [23] effectively encodes continuous positional information in large-scale point clouds by combining voxel hash architecture and local self-attention, reducing quantization error while maintaining consistency with rigid transformations, and balancing inference time and accuracy. In the field of self-attention research, PReFormer [36] proposed a memory-efficient 3-D PCT, which optimizes memory complexity through reversible functions and linearized self-attention mechanisms, while adopting a ∇-shaped architecture to enhance model performance. Compared to traditional methods, it significantly reduces the number of parameters and memory consumption. Additionally, another work [37] introduced the global feature self-attention encoding (GFSAE) module and the weighted semantic mapping (WSM) module. These modules enhance channel-wise attention to global salient features through self-attention mechanisms and incorporate semantic category constraints, effectively improving semantic segmentation performance in large-scale urban street point clouds. These methods provide important references and insights for optimizing memory usage and enhancing segmentation accuracy.

Summarizing the various methodologies, image-based and voxel-based approaches utilize 2-D imaging techniques and 3-D convolutional networks to extract detailed features. Yet, these processes risk losing structural information and can be costly in terms of computation, particularly with high-resolution datasets. Point-based methods, on the other hand, work directly with the raw point cloud data, preserving all information and adeptly capturing both global and local features. Nonetheless, scaling up to large-scale data increases computational complexity and processing time. Attention-based methods shine in their ability to discern complex geometric structures and navigate the sparsity and irregularity of point clouds, focusing on critical features and enhancing performance in analysis tasks. However, the sophistication of these models demands considerable computational and memory resources. In light of the aforementioned insights, this article introduces VEF-Net, a novel deep learning network designed to address the challenges of computational efficiency and feature representation in large-scale point cloud processing. VEF-Net stands out for its innovative approach to reducing computational complexity and improving feature representation, representing a notable advancement for applications such as land cover analysis.

III. Methods

A. Overview

The VEF-Net, as shown in Fig. 1, follows the widely used encoder–decoder framework, augmented with residual connections to enhance learning capabilities. The VEF-Net consists of three essential components: the uniform grid downsampling module, which converts continuous point cloud data into a regular voxel structure to facilitate subsequent processing; the point cloud geometric information enhancement module, designed to expand the feature set derived from voxelized data; and the local–global feature fusion module, which integrates these features for precise semantic segmentation.

B. Uniform Voxelization Downsampling Module

Voxel downsampling is a critical preprocessing step for point cloud data [38]. It converts the continuous point cloud space into a normalized discrete voxel grid, enabling more efficient analysis and processing in subsequent stages. The voxel downsampling process is executed by assigning each point p_i within the point cloud to a predefined voxel cell size, governed by the scaling factor s. This process can be expressed by the following equation:

$$d_i = \left\lfloor \frac{p_i}{s} \right\rfloor \tag{1}$$

where $\lfloor \cdot \rfloor$ denotes the floor operation, and $s \subseteq \mathbb{R}^d$ is the scale factor. Here, it's important to note that a higher value of s

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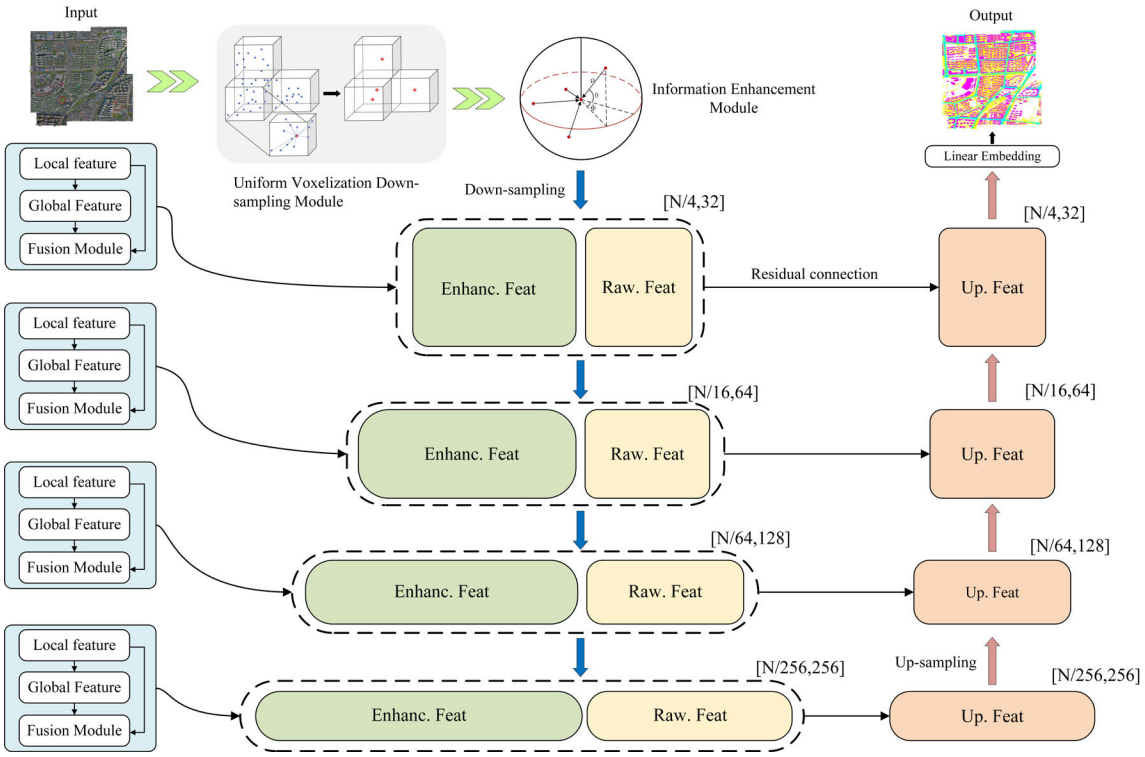


Fig. 1. Overview of the introduced VEF-Net architecture, the raw point cloud is first processed through uniform voxel downsampling to reduce redundancy and complexity. It then passes through the information enhancement module to extract local geometric features. The green parts represent enhanced features, while the dashed boxes show the local–global feature fusion during downsampling. The red parts represent upsampling features, with residual connections preserving the original information. Finally, the network outputs predicted labels for tasks like semantic segmentation or classification, enabling accurate point cloud processing.

equates to a finer resolution of the voxel grid. To maintain the distinctiveness of each voxel cell during downsampling and to curtail memory usage, a hash function h is imposed on each discretized point coordinate d_i. The resulting hash values are then sorted, duplicates are eliminated, and the final set of unique hash values p' is defined by the following formula:

$$p' = \text{unique}(\text{sort}(h(d_i)) \,|\, \{p_i \subseteq P\}). \tag{2}$$

Each element in the set p' represents a distinct voxel location within the point set p. The elimination of duplicate hash values guarantees that the voxel downsampling outcome is a concise and nonredundant representation, thereby substantially reducing the computational requirements. Section V will present an experimental comparison of the impact that different scale factors s on the efficiency of voxel downsampling and data accuracy. This will aid in pinpointing the optimal voxel downsampling parameters that expedite the processing of point clouds, while also maximizing the retention of the features inherent in the original data.

C. Local Geometric Information Enhancement Module

The purpose of this module is to enhance the final semantic segmentation by enriching the local geometric information of the point cloud. This is accomplished by employing the KNNs algorithm to identify the K closest neighbors for each point. Utilizing these neighbors, the module then computes local geometric features of the point cloud, thereby enriching its feature set.

Drawing insights from prior studies [15], [39], let $p' = \{p'_1, \ldots, p'_n\} \subseteq \mathbb{R}^{n\times 3}$, where $P'_i = \{p'_{i1}, \ldots, p'_{ik}\}$ represents the neighboring points for P'_i. To enhance the representation of local structure and geometry, the module transforms the local point cloud from the Cartesian coordinate system (x, y, z) into a spherical coordinate system (ρ, θ, ϕ). The computation process can be expressed as

$$\rho = \sqrt{x^2 + y^2 + z^2} \tag{3}$$

$$\theta = \arccos\left(\frac{z}{\rho}\right) \tag{4}$$

$$\phi = \text{arctan2}(y, x) \tag{5}$$

where ρ represents the Euclidean distance from neighboring points to the central point, θ denotes the polar angle, and ϕ represents the azimuthal angle. The K neighboring points are sorted in a counterclockwise direction based on the azimuthal angle ϕ.

By identifying two adjacent points p'_{ij} and p'_{ij+1} relative to p'_i, the modal calculates the spatial centroid $c_i = \text{mean}(p'_i, p'_{ij}, p'_{ij+1})$ and determines its spatial normal vector v_k.

To maintain uniformity in the direction of the normal vectors and prevent geometric discrepancies, a masking function pos_mask is introduced to confirm the positivity of each component. The spatial normal vector is defined as

$$v_k = \text{pos_mask}\big(\text{cross}\big(p'_{ij} - p'_i, p'_{ij+1} - p'_i\big)\big). \tag{6}$$

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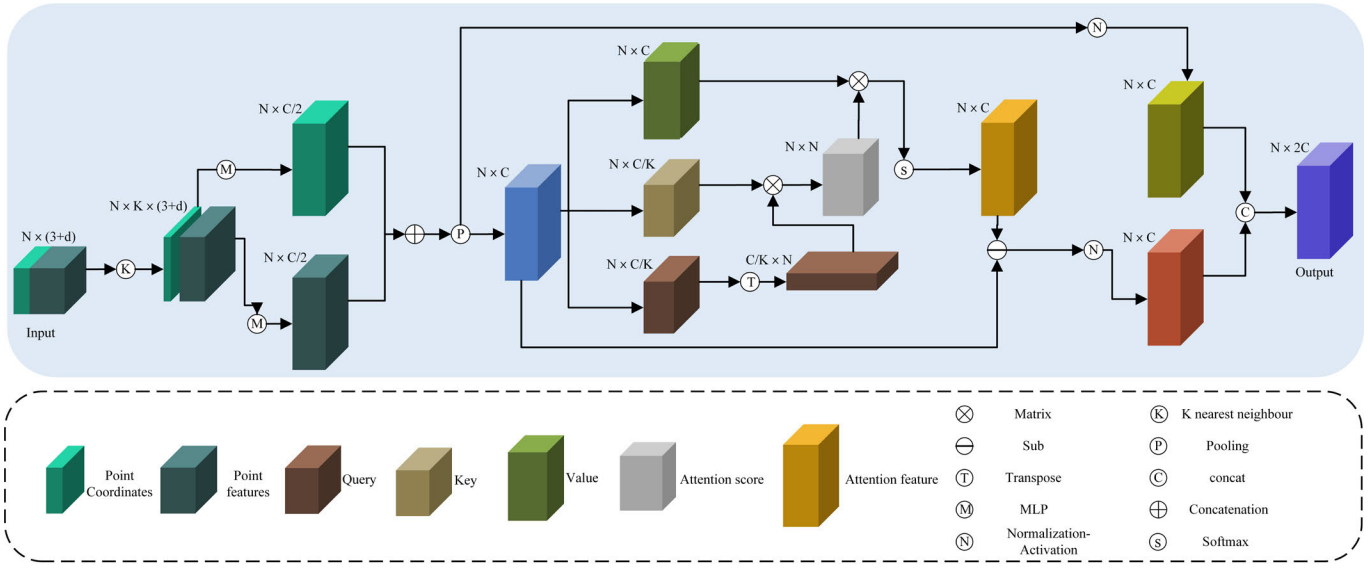


Fig. 2. Overview of the introduced local–global feature fusion architecture, the input point cloud is first processed through KNN to extract local features by finding the neighbors of each point, capturing the geometric relationships in its local region. Then, the model generates query, key, and value matrices, and calculates attention scores through a self-attention mechanism to extract global geometric features. Finally, local and global features are fused together via residual connections and feature concatenation. This fusion ensures that the model captures both local details and global semantics of the point cloud, enabling more accurate point cloud classification or segmentation.

Subsequently, by utilizing $K/2$ spatial centroids along with their corresponding spatial normal vectors, the module then computes the spatial geometric position of the point cloud as $X = \text{sum}(c_i, v_k)/\sqrt{3}$. The geometric features extracted from the point cloud are concatenated, thus enhancing its spatial geometric properties. Additionally, a trainable MLP function is incorporated to further elaborate on the point cloud's features.

D. Local–Global Feature Fusion Module

The local–global feature fusion module, as shown in Fig. 2, captures comprehensive information from the data by combining global and local features. Global features represent the overall information, equipping the model with a holistic view of the dataset, whereas local features concentrate on the finer details and variations, increasing the model's sensitivity to subtle changes. The complementary combination of global and local features facilitates a more exhaustive understanding of the nuances present in point cloud information.

1) Local Feature Fusion: This component is designed to thoroughly extract local features from the point cloud by utilizing the KNNs algorithm to identify K surrounding points for each point, resulting in a feature tensor of size $[N, K, C]$. Here, N denotes the total number of points, K signifies the number of neighbors per point, and C indicates the dimensionality of each point's features. During the feature extraction phase, the spatial coordinates and other attributes of the point cloud are treated separately. For large-scale geospatial point cloud data, managing GPU memory is crucial. By separating the point cloud's spatial information from other features, significant memory consumption can be reduced, effectively preventing out-of-memory errors. This approach not only optimizes the allocation of computational resources but also enhances memory usage efficiency. The coordinate data undergo 1-D convolutional operations coupled with batch normalization to capture the spatial geometry of the point cloud. Other feature types are processed through analogous networks to seize supplementary characteristics. To integrate both spatial and nonspatial information and to deeply mine local features, a multilayer network is established, each consisting of a linear transformation, normalization, and nonlinear activation, culminating in an MLP. Within the MLP, spatial and nonspatial features are merged, bolstering the model's capacity to represent intricate local structures. This layered architecture empowers the model to acquire a rich set of features that encapsulate the attributes of each point and its vicinity.

2) Global Feature Fusion: While local features are pivotal for detailing point clouds, they may fall short in representing the global structure and contextual information, which is vital for distinguishing objects with similar local features but distinct global forms. Furthermore, relying solely on local features can be inadequate for deciphering complex scenes. To overcome these limitations and deepen the model's comprehension of 3-D point clouds, a self-attention mechanism is deployed for global feature integration. VEF-Net combines convolution and self-attention mechanisms. Traditional fully connected layers (such as those used in calculating queries, keys, and values) result in a computational complexity of $O(N^2 * C)$, where N is the number of points and C is the feature dimension.

To address this issue, this article replaces fully connected layers with convolution operations, reducing the computational complexity from $O(N^2 * C)$ to $O(N * K * C)$, where K is the size of the convolution kernel, and K is much smaller than N. This improvement not only enhances computational efficiency but also boosts the performance of global feature aggregation, effectively reducing both computational load and memory consumption. This mechanism adeptly seizes the global interconnections among points within the point cloud by conducting self-attention operations on the aggregated

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feature tensor $[N, C]$, thus internalizing the broader contextual framework.

Inspired by previous research [19], [20], the global self-attention mechanism leverages the self-attention approach from the query matrix $\boldsymbol{Q}$, key matrix $\boldsymbol{K}$, and value matrix $\boldsymbol{V}$. To optimize computation, convolutional weights are shared between the query $\boldsymbol{Q}$ and key $\boldsymbol{K}$ matrices. This weight-sharing strategy reduces redundant computations and storage overhead. Specifically, the same convolution kernel is used to process both the query and key matrices, eliminating the need to independently store and compute different weight matrices during each calculation, thus effectively reducing memory requirements and computational load. The input matrix $\boldsymbol{X}$ is of size $\boldsymbol{X} \in \mathbb{R}^{N \times C}$, where N denotes the number of points and C represents the feature dimension. The computation process is articulated in the following formula:

$$\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} = \mathbf{Conv1D}(\boldsymbol{X})\left(\boldsymbol{W}^{\boldsymbol{Q}}, \boldsymbol{W}^{\boldsymbol{K}}, \boldsymbol{W}^{\boldsymbol{V}}\right) \tag{7}$$

where $\boldsymbol{W}^{\boldsymbol{Q}}$, $\boldsymbol{W}^{\boldsymbol{K}}$, and $\boldsymbol{W}^{\boldsymbol{V}}$ are corresponding weight matrices, with $\boldsymbol{W}^{\boldsymbol{Q}}$ and $\boldsymbol{W}^{\boldsymbol{K}}$ are being shared to economize computational resources. Where $\boldsymbol{S}$ in the following formula represents the attention scores matrix, which quantifies the relationships between all points:

$$\boldsymbol{S} = \boldsymbol{Q} \cdot \boldsymbol{K}^{\boldsymbol{T}} \in \mathbb{R}^{\boldsymbol{N} \times \boldsymbol{N}}. \tag{8}$$

The self-attention scores are calculated from the original feature matrix X to assess the similarity between different points. The self-attention score is derived from the dot product of the query matrix $\boldsymbol{Q}$ and the key matrix $\boldsymbol{K}$, followed by normalization using the factor $\sqrt{d_k}$ and the softmax function ρ to convert it into attention weights.This approach ensures numerical stability and prevents gradients from becoming too small. Once the weights are applied to the value matrix V, a matrix of weighted features is generated. This matrix is subsequently integrated with the original feature matrix $\boldsymbol{X}$, forming a residual connection. The residual connection helps maintain the flow of features, avoiding information loss in deep networks. Upon completion of normalization and activation operations, the output feature matrix $\boldsymbol{X}'$ in the following formula, which shares the same dimensions as the original feature matrix, introduces nonlinearity and further stabilizing model training:

$$\boldsymbol{X}' = \boldsymbol{\varphi}\left(\boldsymbol{\rho}\left(\frac{\boldsymbol{S}}{\sqrt{\boldsymbol{d_k}}}\right) \cdot \boldsymbol{V} + \boldsymbol{X}\right). \tag{9}$$

A detailed depiction of the self-attention module is provided in Fig. 2, illustrating the transformation of input features and the aggregation of weighted features to generate output features. By sharing the weights and bias parameters across queries and keys, and adopting small convolution kernels, the model significantly streamlines its parameter count and computational demands, thus boosting its efficiency and responsiveness. In the local–global feature fusion module, the global feature extraction is strategically placed after the local feature extraction. This design sequence is based on two key considerations: first, the sufficient extraction of local features provides a fine-grained semantic foundation for global relation modeling; second, this order effectively prevents premature global aggregation, thus avoiding the loss of details. Specifically, after the completion of local feature learning, the system performs downsampling on the local feature matrix using max pooling, reducing the number of points in the point cloud from N original points to M center points ($M = N/r$, with $r = 4$ as the downsampling rate). This operation relies on two core mechanisms: 1) spatial dimension compression: effectively reducing the spatial dimension through the downsampling rate r and 2) receptive field expansion: each center point's spherical region contains $K = 32$ original point features, which is akin to expanding the receptive field of a single point from the local scale to the global scene scale. After global feature learning, a simple dual-route linear concatenation fusion method is used. Ablation experiments have shown that this method demonstrates significant performance improvements.

IV. Experiment

A. Experimental Setup

1) Experimental Environment: The experiments were conducted on a computer equipped with an NVIDIA GTX 4090 GPU and an Intel[1] Xeon[1] Silver 4210 CPU at 2.20 GHz. The primary development environment utilized was Python 3.8, in conjunction with PyTorch 1.8.0 + cu111.

2) Experimental Evaluation Metrics: The evaluation of model accuracy was conducted using mean intersection over union (mIoU), overall accuracy (OA), and mean accuracy per Class (mAcc) as the principal metrics. mIoU measures the average overlap between the model's predicted segmentation and the ground truth by calculating the ratio of the intersection area to the union area of predicted and actual labels. OA indicates the proportion of correctly classified points out of the total points predicted, while mAcc refers to the proportion of correctly identified points for each class against the total points in that class. The metrics are mathematically defined by the following formulas:

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i + \text{FN}_i} \tag{10}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}} \tag{11}$$

$$\text{mAcc} = \frac{1}{N}\sum_{i=1}^{N}\frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}. \tag{12}$$

Additionally, the model's complexity is gauged by the number of parameters (Params), and its operational efficiency by the number of floating point operations per second (FLOPs), which quantifies the computational load during a single forward pass of the model. These metrics are pivotal for assessing the model's performance and resource requirements.

B. Experimental Dataset

The experimental evaluation was conducted utilizing two primary datasets: the proprietary Bengbu dataset and the publicly accessible Toronto-3D dataset. The Bengbu dataset was generated through autonomous aerial vehicle (AAV)

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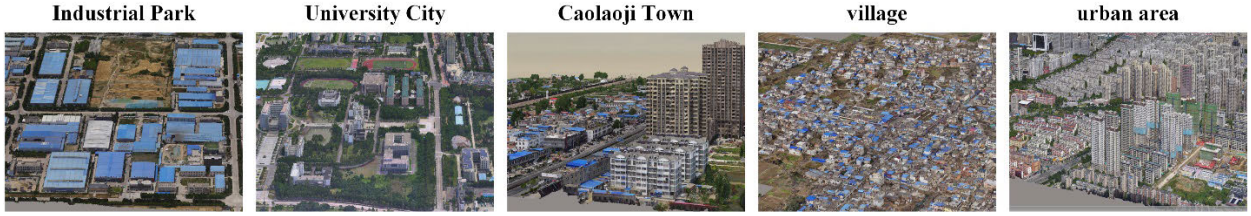


Fig. 3. Illustration of five distinct scenes captured from the Bengbu oblique photography point cloud dataset.

TABLE I
BENGBU DATASET PARAMETERS

area	Industrial park	University City	Caolaoji Town	Village	Urban area
Number of points (1×10^7)	8.1	10	3.5	3.5	13
Average height (m)	54.9	134.9	104.9	38.1	148.6
Area (km^2)	2.5	3.0	1.1	2.6	3.9
Points per m^2	32	34	33	33	33

oblique photogrammetry and digital terrain vector map, effectively capturing the diverse urban landscapes of Bengbu city. As shown in Fig. 3, this dataset is characterized by its intricate 3-D spatial information and RGB color data, making it highly suitable for semantic segmentation tasks in complex urban environments.

The Bengbu dataset encompasses a variety of five distinct scenes: an Industrial Park, University City, Caolaoji Town, a Village, and an Urban Area. Each scene is replete with nuanced geographical and structural details. Table I presents a detailed account of statistical parameters for these regions, including the number of points, average elevation, spatial extent, and points per square meter. The dataset's comprehensive diversity, dense point distribution, and broad geographical scope render it exceptionally suitable for undertaking semantic segmentation of urban object point clouds on a large scale.

The data acquisition initiative commenced with the downsampling of extensive 3-D point clouds, a strategic step to optimize the balance between data manageability and the preservation of essential information. Utilizing an integrated approach of structure from motion (SfM) and multiview stereo (MVS) techniques, drones, specifically the Guangzhou Hi-Target iFyL V5 models equipped with iCam-Q5 mini cameras, flew predefined routes to capture high-fidelity images of urban and rural areas in Bengbu. These images were then subjected to SfM-MVS analysis using DP-Smart software, resulting in a coherent 3-D structural model. Further, the dataset underwent meticulous semantic annotation, laying a robust groundwork for AI model training and the establishment of precise accuracy metrics. For a more exhaustive delve into the data specifics, please refer to the detailed descriptions provided in [21], [40].

The Toronto-3D dataset [41], crafted for 3-D urban environment comprehension and semantic segmentation, was assembled by a team of researchers from the University of Toronto. It comprises real urban scenes from downtown Toronto, with data acquisition performed using vehicle-mounted LiDAR sensors to capture comprehensive 3-D point cloud representations of the urban landscape. Each point cloud is annotated with 3-D coordinates (x, y, z), reflection intensity, and granular semantic labels encompassing a variety of urban elements such as buildings, roads, vegetation, vehicles, and pedestrians. This richly annotated dataset facilitates the execution of 3-D semantic segmentation across a spectrum of urban settings.

C. Bengbu Dataset Experiment

To validate the performance of VEF-Net, samples encompassing buildings, natural environments, and roads from various regions of the Bengbu dataset were selected for semantic segmentation, with the results depicted in Fig. 4. To ensure greater representativeness, Fig. 4 presents a visual comparison of semantic segmentation outcomes across diverse scenes for a selection of state-of-the-art models—PointNet, PointNet++, RepSurf, and PT—as well as the proposed VEF-Net. The comparative analysis, as illustrated across Fig. 4(a)–(f), clearly indicates VEF-Net's superior performance in segmenting these diverse urban and natural elements. VEF-Net particularly excels in the precise delineation of boundaries, especially where roads intersect with buildings. Table II presents the quantitative experimental results of different models on the Bengbu dataset. To ensure a fair comparison and verify the models' ability to extract semantic features, we compared multiple mainstream models based on metrics such as mIoU, overall OA, mAcc, parameter count, FLOPs, training time, and inference time.

While RepSurf outperforms in evaluation metrics as indicated in Table II, it has shown a tendency to misclassify roads excessively. The PT, noted as PT in Fig. 4, exhibits enhanced semantic segmentation accuracy in specific scenarios but suffers from significant misclassifications in Fig. 4(a). Additionally, the point transformer's training and inference times, as indicated in Table II, are considerably longer than other methods, potentially impacting its practical application in semantic segmentation tasks.

Compared to PointNet and PointNet++, which display considerable misclassifications in certain areas as illustrated in Fig. 4(b) and (c), VEF-Net provides more distinct and

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TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE BENGBU DATASET

Method	mIoU	Bldg.	Rd.	Veg.	OA	mAcc	Params	FLOPs /1000	Train times/H	Infer times/M
Pointnet[12]	91.6	95.4	86.0	93.5	96.7	94.3	0.97	1.16	11.98	18.22
Pointnet++(msg)[13]	92.4	94.6	88.9	93.6	96.9	97.2	1.88	1.30	13.67	20.30
PCT[20]	88.9	93.2	82.2	91.4	94.6	93.2	2.87	2.14	29.89	37.73
FastPT(smaller)[23]	88.2	92.5	81.1	91.0	93.7	92.3	10.80	-	9.46	13.35
RandLA-Net[22]	92.2	97.3	84.9	94.4	96.1	95.7	1.31	-	12.76	18.67
Repsurf[15]	90.1	94.9	83.7	91.6	95.8	96.4	0.98	1.04	6.79	9.78
Kpc[42]	89.3	93.3	82.2	92.4	94.7	92.5	14.30	-	18.77	26.53
PT[19]	92.7	97.0	85.7	95.5	96.2	97.6	7.78	35.9	50.87	140.35
DU[43]	92.9	96.2	89.4	93.1	97.4	96.5	8.10	-	13.88	43.38
VEF-Net(Ours)	91.4	94.7	86.8	92.6	94.9	95.7	1.08	1.14	4.08	8.33

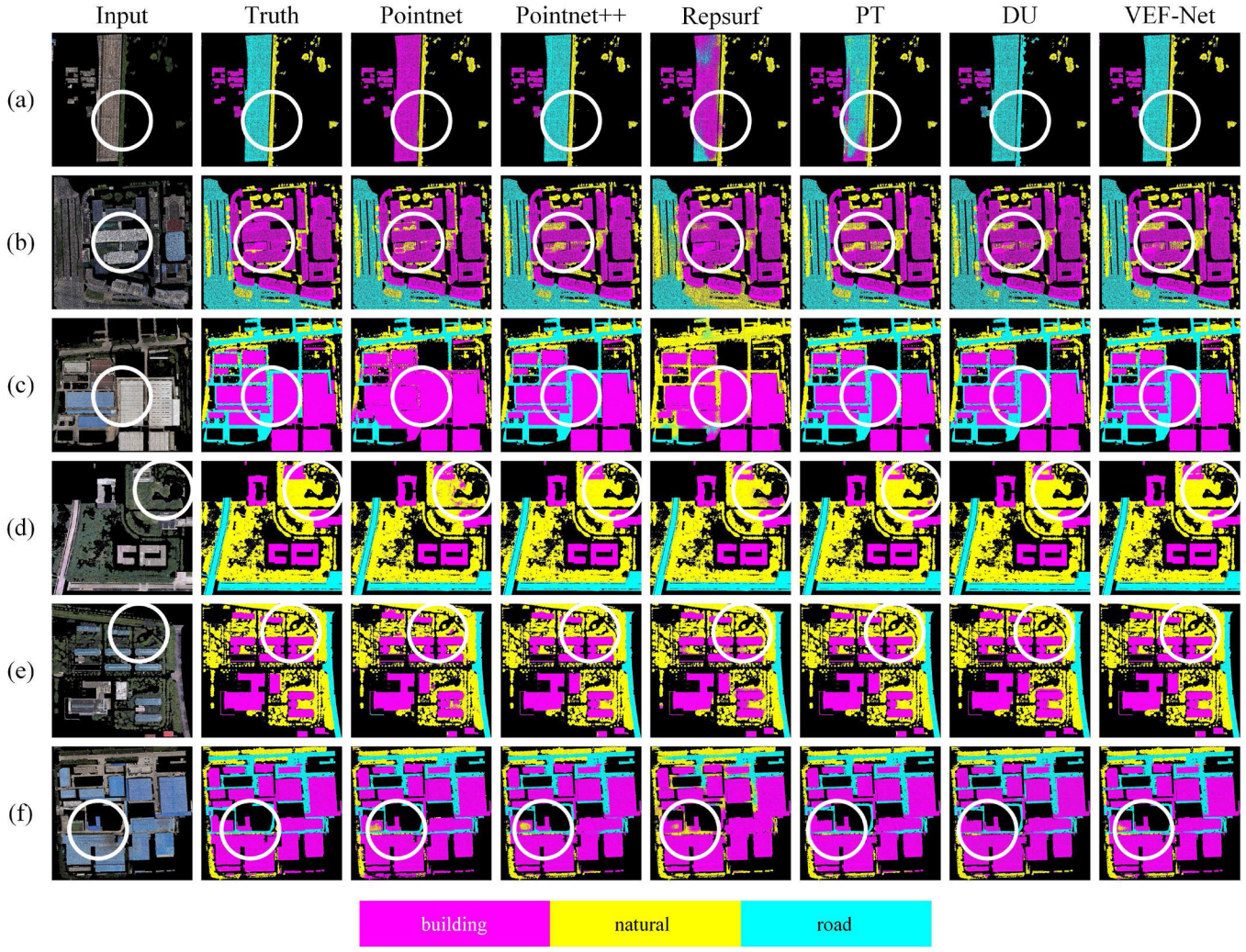


Fig. 4. Visualization comparison of test results from mainstream models on the Bengbu dataset. This figure illustrates the performance of six different models—(a)–(f) PointNet, PointNet++, Repsurf, Transformer, and VEF-Net—across five distinct regions in the Bengbu dataset. The colors represent different classes: buildings in magenta, natural areas in yellow, and roads in cyan. The white circle in the figure highlights the significant differences in segmentation between VEF net and different mainstream models.

accurate segmentation edges. While PointNet encounters challenges in differentiating between buildings and their adjacent natural environments, VEF-Net maintains clear and well-defined boundaries. Additionally, VEF-Net's segmentation of

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intricate details such as road markings and building structures, as shown in Fig. 4(d) and (f), is characterized by its ability to capture local details with precision while upholding the overall structural coherence. This enhanced performance is primarily due to VEF-Net's local–global feature fusion module, which effectively integrates detailed local geometric information with a comprehensive global context. This integration not only refines the model's sensitivity to fine-grained features but also strengthens its capacity to understand and process the broader spatial relationships within the point cloud data, thus optimizing VEF-Net's handling of complex segmentation tasks.

As shown in Table II, the performance of VEF net in specific categories is differentiated. For example, in the road (Rd.) classification, VEF net reaches 86.8%, which is better than Pt (85.7%) and randla net (84.9%), indicating that it has a strong ability to capture road structures with similar elevations. In the classification of buildings (bldg.), VEF net is slightly higher than PointNet++(94.6%) with 94.7% accuracy, but lower than Pt (97.0%), randla net (97.3%), and diffusion unit (DU) (96.2%), reflecting that there is still room for optimization in dense building areas. In the classification of vegetation (veg.), 92.6% of VEF net is slightly inferior to PointNet++(93.6%) and Pt (95.5%), which may be limited by the detailed modeling ability of complex vegetation scenes. It is worth noting that VEF-Net demonstrates a significant advantage in computational efficiency. Although VEF-Net's mIoU of 91.4% is marginally lower than PT's 92.7% and PointNet++'s 92.4%, it excels in minimizing parameter usage and accelerating inference. VEF-Net is defined by a modest parameter count of 1.08 M and a computational complexity of 1.14G FLOPs, positioning it favorably against other approaches. For example, PCT requires 2.87 M parameters and 2.14G FLOPs, whereas PT is significantly more demanding with 7.78 M parameters and a substantial 35.9G FLOPs. Additionally, VEF-Net's proficiency in training and inference is commendable, necessitating only 4.08 h and 8.33 min, respectively, thus outpacing other models in these metrics. These advantages highlight VEF-Net's readiness for practical applications, particularly where there is a need to balance high segmentation accuracy with rapid training and inference processes, such as in point cloud semantic segmentation tasks with a smaller number of categories. This makes VEF-Net a strong candidate for real-world scenarios where efficiency is paramount.

D. Validation on Public Dataset

Having demonstrated remarkable efficiency and precision in the Bengbu dataset, which encompasses a limited set of categories, VEF-Net's capabilities were further assessed for multicategory land cover segmentation using the. Toronto-3D dataset. The evaluation was conducted using only the positional data from the dataset as input features for the point clouds to ensure impartiality. Table III, derived from the official Toronto-3D GitHub repository, presents a comprehensive comparison of various models' performance on the Toronto-3D dataset, with a particular focus on the distinguished results achieved by VEF-Net across metrics such as mIoU, OA, and class-specific IoU.

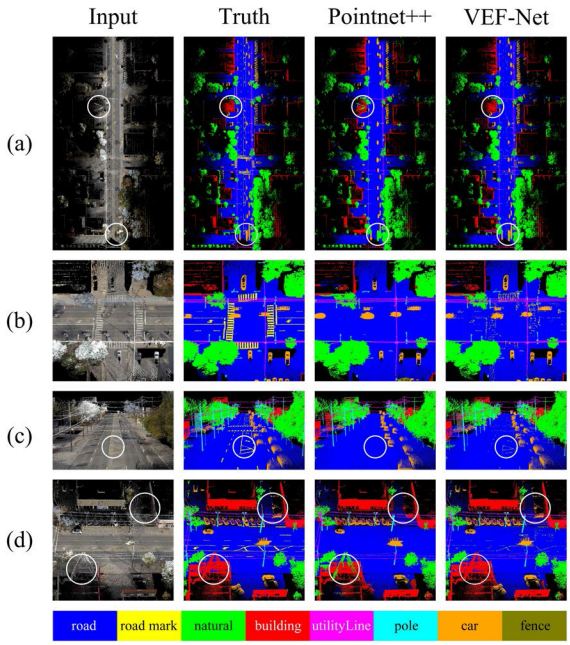


Fig. 5. Test results on the Toronto-3D dataset are visually compared. (a) Overall visualization of the Toronto-3D data, providing a comprehensive view of the scene. (b)–(d) Enlarged views of different areas or angles to highlight detailed segmentation performance. The figure focuses on key semantic classes such as roads, road markings, natural areas, buildings, utility lines, poles, cars, and fences. Different colors represent these categories: roads are blue, road markings are yellow, natural areas are green, buildings are red, utility lines are purple, poles are cyan, cars are orange, and fences are brown. The white circles in the figure highlight the significant differences in segmentation details between VEF-Net and PointNet++. VEF-Net is able to more accurately identify and distinguish the boundaries of different categories, while PointNet++ exhibits more blurred boundaries and misclassifications in certain complex areas.

For visual comparison of the segmentation results, the widely recognized PointNet++ model was selected as a benchmark for comparison, with the dataset's official website providing the comparison results shown in Table III. Fig. 5 illustrates the comparative segmentation results of VEF-Net and PointNet++. Fig. 5(a) offers a panoramic view of the L002 scene, while Fig. 5(b)–(d) focuses on specific regions and vantage points to emphasize the detailed segmentation performance. The white circles in the figures indicate areas where VEF-Net exhibits superior performance compared to PointNet++.

The visual segmentation comparison indicates that VEF-Net demonstrates a more nuanced capability in the detailed delineation of road markings, natural landscapes, buildings, and vehicles in comparison to PointNet++. Particularly in Fig. 5(b) and (d), the network's ability to resolve subtleties in these categories is more pronounced. Consistent with the results obtained from the Bengbu dataset, VEF-Net not only accurately identifies object boundaries but also maintains consistent classification across various regions. In contrast, PointNet++ shows some limitations, particularly in areas with complex details, where it can exhibit inaccuracies and confusion. Notably, VEF-Net's segmentation capabilities are further demonstrated in Fig. 5(c) and (d), particularly for road markings and fences.

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TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE TORONTO-3D DATASET [41]

Method	mIoU	OA	IoUs							
			road	road m.	natural	build	Util l.	pole	car	fence
Pointnet++[13]	59.5	92.6	92.9	0.0	86.1	82.1	60.9	62.8	76.4	14.4
MS-PCNN[44]	65.9	90.0	93.8	3.8	93.4	82.5	67.8	71.9	91.1	22.5
MS-TGNet[41]	70.5	95.7	94.4	17.1	95.7	88.8	76.0	73.9	94.2	23.6
KPFCNN[42]	69.1	95.4	94.6	0.1	96.0	91.5	87.6	81.5	85.6	15.7
TGNet[45]	61.3	94.1	93.5	0.0	90.8	81.6	65.3	62.9	88.7	7.9
RandLA-Net	74.1	94.9	93.5	12.5	96.6	92.3	86.2	80.1	88.0	43.5
DGCNN[46]	61.8	94.2	93.8	0.0	91.2	80.3	62.4	62.3	88.2	15.8
VEF-Net(Ours)	68.4	93.8	93.1	20.4	92.5	91.0	74.2	64.2	85.3	26.5

The quantitative outcomes presented in Table III emphasize VEF-Net's superior performance across overall metrics of mIoU and OA, particularly excelling in the intricacies of fine-grained segmentation tasks. In multiclass land cover segmentation tasks, road markings are often one of the most challenging categories due to their small area and detailed complexity. A model's proficiency in this area is indicative of its adeptness at capturing subtle details while maintaining awareness of the encompassing context. As shown in Table III, VEF-Net performs excellently with an mIoU of 68.4% and an OA of 93.8%. In the road marking category, VEF-Net significantly outperforms other methods with an mIoU of 20.4%. Additionally, VEF-Net achieves an mIoU of 26.5% in the fence category, 64.2% in the pole category, and 91.0% in the building category, further demonstrating its effectiveness in multiclass, fine-grained land cover segmentation tasks.

Benefiting from its local–global feature fusion, VEF-Net achieved the highest IoU score in the road markings category, reaching 20.4%, and thus significantly outperformed other mainstream models. In comparison, models such as PointNet++, dynamic graph CNN (DGCNN), Taylor Gaussian mixture model network (TGNet), and multi-scale point-wise convolutional neural networks (MS-PCNN) struggled in this category, frequently failing to accurately delineate the intricate road markings. The architecture of PointNet++, which emphasizes local point features without fully integrating global context, leads to significant inaccuracies when segmenting areas with rich detail and complex geometry, like road markings. Likewise, DGCNN and TGNet, despite their success in other categories, encounter difficulties in achieving precise segmentation for finer details, such as road markings. RandLA-Net, despite its comparable performance to VEF-Net in segmenting natural environments and buildings and a slight edge in OA, requires significantly greater computational resources and processing time. Additionally, its complex data preprocessing demands can limit its practicality in resource-limited settings.

Experiments conducted on the public Toronto-3D dataset demonstrate that VEF-Net matches the precision of current mainstream models in overall mIoU and OA. Furthermore, VEF-Net exhibits outstanding capabilities in fine-grained segmentation across various categories. Its top score in the road marking category, along with high rankings in the fence, pole, and building categories, attests to its efficacy in multicategory land cover classification, underscoring its extensive applicability in practical scenarios.

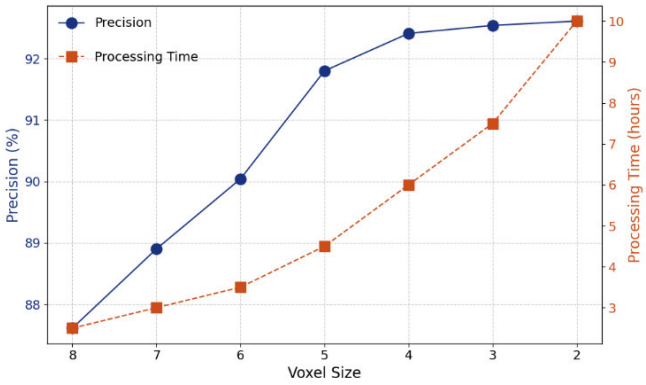


Fig. 6. Accuracy and training duration under different voxel resolutions.

V. ABLATION STUDY

A. Accuracy and Efficiency Under Different Voxel Sizes

This section explores the impact of voxel downsampling across a range of voxel sizes on the training time and segmentation accuracy of VEF-Net, utilizing the Bengbu dataset. The study evaluates voxel sizes from 8 to 2, covering various spatial resolutions to assess the influence of voxel granularity on the model's learning efficiency and prediction accuracy. To optimize performance and hardware resource utilization, GPU usage was carefully regulated within the 65%–80% range by adjusting parameters such as batch size, based on preliminary experimental outcomes. This strategy was essential for pinpointing the optimal balance between high model performance and resource conservation.

Fig. 6 demonstrates that the model's training time increases in proportion to the decrease in voxel size. This indicates a direct relationship where finer voxel sizes, while enhancing the detail of the model's input, also demand more computational resources and thus extend the training time. On the other hand, the segmentation accuracy of the model improves with smaller voxel sizes until it reaches a plateau or slightly decreases at the smallest sizes. This trend underscores the importance of balancing training efficiency with segmentation accuracy when determining voxel size.

Larger voxel sizes may speed up training but could diminish accuracy, while smaller sizes, despite enhancing accuracy, can lead to higher computational expenses. Identifying this balance

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TABLE IV
IMPACT OF LOCAL GEOMETRIC INFORMATION ENHANCEMENT COMPONENTS ON MODEL PERFORMANCE

Components	Centroid c_i	Vector v_k	Position $\mathbf{X}$	mIoU
Local Geometric Information Enhancement				89.3
	√			90.2
	√	√		90.6
	√	√	√	91.4

TABLE V
IMPACT OF THE NUMBER OF NEIGHBORING POINTS (K) ON MODEL PERFORMANCE

K	4	8	16	32
mIoU	89.8	91.4	91.0	90.6

is crucial for achieving high model performance without incurring unnecessary computational costs.

B. Effectiveness of Local Geometric Information Enhancement Module

The impact of varying point cloud feature components on the performance of VEF-Net was scrutinized through ablation experiments on the Bengbu dataset, as shown in Table IV. The results demonstrate a consistent enhancement in the model's mIoU as more geometric features are incorporated. With the use of the centroid c_i alone, the mIoU stands at 89.3%. The addition of the vector v_k further improves the mIoU to 90.2%. When all three features, the centroid c_i, vector v_k, and position $\mathbf{X}$, are integrated, the mIoU achieves its maximum value of 91.4%. This upward trend underscores the substantial benefits of enriched local geometric information in bolstering the model's segmentation accuracy.

Further analysis focused on the impact of the parameter K, which denotes the number of neighboring points, on the model's performance, as illustrated in Table V. The outcomes demonstrate that raising K does not invariably lead to higher segmentation accuracy. While a lower K value might restrict the availability of local information, it also avoids the incorporation of extraneous points that could diminish the model's accuracy in distinguishing features. Conversely, a higher K value risks including irrelevant points that are far from the sampling point, potentially undermining the model's capacity for precise segmentation. The experimental results indicate that enriching local geometric information and carefully selecting the K value is key to balancing detail and relevance, enhancing the model's segmentation accuracy.

C. Impact of Complexity and Local–Global Feature Fusion

The ablative experiment validated the importance of the local–global feature fusion module and the network complexity optimization strategy. As shown in Table VI, removing this module reduced the model's parameter count to 0.80 M, but the mIoU significantly dropped from 91.4% to 86.2%, indicating the crucial role of this module in improving performance. In the comparison of single-feature extraction, using only local features resulted in an mIoU of 89.3%, using only global features yielded an mIoU of 88.7%, while using only the global self-attention mechanism achieved an mIoU of 90.7%. These results demonstrate that the fusion of local and global features is essential for performance enhancement. Theoretically, local features capture spatial information, while global features provide global contextual information, and their complementarity improves the model's representational capability, thus enhancing overall performance. Comparing feature fusion strategies, it was found that the linear concatenation strategy outperforms the weighted average fusion in terms of performance with fewer parameters. Theoretically, linear concatenation effectively reduces redundant computations without sacrificing information, maintaining higher efficiency compared to weighted average fusion. When adjusting the self-attention channel dimensions, reducing the number of channels to 1/8 of the original value resulted in a parameter count of 1.06 M and an mIoU of 91.0%. This result is comparable to the full module's 1/4 channel configuration but with lower complexity. However, when the number of channels was doubled, the mIoU dropped to 90.1%. This result verifies the nonlinear relationship between model complexity and performance: moderately reducing complexity can improve efficiency without sacrificing accuracy, while excessively high complexity may lead to performance degradation. It also indicates that there exists an "optimal complexity" range, and increasing complexity beyond this range does not lead to performance gains but may instead cause performance decline. Notably, when the original channel configuration was maintained, the model's mIoU was 91.2%, only 0.2% lower than the full module's 91.4%, validating the balance between efficiency and performance in this configuration. In conclusion, local–global feature fusion is the core driving force of the model's performance, and by adjusting network complexity, the computation can be reduced effectively while maintaining accuracy.

TABLE VI
EFFECTIVENESS OF THE LOCAL–GLOBAL FEATURE FUSION MODULE

Configuration	Params	mIoU
Remove Local-Global Feature Fusion Module	0.80	86.2
Retain Only Local Features	0.90	89.3
Retain Only Global Features	0.97	88.7
Retain only Global Self-Attention Mechanism	0.99	90.7
Replace the Fusion Method with the Weighted Average	1.46	91.0
Change the Attention Channel Features to the original 1/8	1.06	91.0
Change the Attention Channel Features to twice the original one	1.24	90.5
Keep the Attention Channel Features unchanged	1.15	91.2
All Modules	1.08	91.4

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VI. Conclusion

The VEF-Net, as detailed in this article, represents a thoughtful advancement in the semantic segmentation of point cloud data. Its architecture, which harmoniously combines local and global feature fusion with voxel downsampling, not only enhances the network's ability to capture fine details and broader context but also significantly improves operational efficiency. This design consideration is particularly beneficial for handling large-scale point cloud datasets, which are common in urban planning and environmental analysis.

Empirical evaluations on both on the proprietary Bengbu dataset and the publicly accessible Toronto-3D dataset have shown that VEF-Net while maintaining comparable accuracy to mainstream models, exhibits a significantly lower parameter count and higher training and inference efficiency. Compared to the top-performing PT model, VEF-Net shows only a slight 1.3% decrease in mIoU. However, it achieves a substantial reduction in parameter count by a factor of 7.2, enhances training efficiency by 12 times, and improves inference efficiency by 16.8 times. Furthermore, the integration of the local–global feature fusion module enhances VEF-Net's performance, particularly in fine-grained segmentation across various categories. This is evident in the Toronto-3D dataset, where VEF-Net achieves a 20.4% accuracy in the road marking category, a result that surpasses other state-of-the-art methods. These findings underscore the efficiency, effectiveness, and robustness of VEF-Net for the segmentation of land cover point clouds.

In future research, algorithms for point cloud feature extraction will be refined to improve accuracy and efficiency in the semantic segmentation of large-scale point clouds. Efforts will also be directed toward reducing the number of parameters and hardware requirements for deep learning models, which is essential for extending the practical applications of deep learning in point cloud segmentation.

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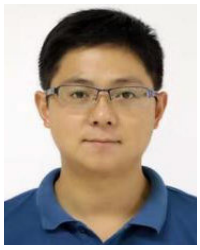
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